Linear Algebra
Complete Notes

BS-IV Mathematics

SALU Khairpur — 2025–2026

Topics Covered

- Matrix Operations ($+, -, \times$)
- Determinants
- Singular & Non-Singular Matrix
- Minors & Cofactors
- Gauss–Jordan Elimination
- Gauss Elimination
- Cramer's Rule
- Orthogonal & Orthonormal Sets
- Projection of a Vector
- Group Theory

1. Matrix Operations

Definition — Matrix

A **matrix** is a rectangular array of numbers arranged in rows and columns. An $m \times n$ matrix has m rows and n columns. A 3×3 matrix:

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

Addition and Subtraction

> Two matrices of the **same order** are added/subtracted element-by-element: $(A \pm B)_{ij} = a_{ij} \pm b_{ij}$.

Example. Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$, $B = \begin{pmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{pmatrix}$.

$$A + B = \begin{pmatrix} 10 & 10 & 10 \\ 10 & 10 & 10 \\ 10 & 10 & 10 \end{pmatrix}, \qquad A - B = \begin{pmatrix} -8 & -6 & -4 \\ -2 & 0 & 2 \\ 4 & 6 & 8 \end{pmatrix}$$

Matrix Multiplication

Definition — Matrix Product

For A $(m \times n)$ and B $(n \times p)$, the product $C = AB$ is $m \times p$ with

$$c_{ij} = \sum_{k=1}^{n} a_{ik}\, b_{kj}.$$

Matrix multiplication is **not commutative** in general: $AB \neq BA$.

Example.

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 0 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ 0 & 3 \\ 1 & 0 \end{pmatrix}$$

$$AB = \begin{pmatrix} 1\cdot 2 + 2\cdot 0 + 0\cdot 1 & 1\cdot 1 + 2\cdot 3 + 0\cdot 0 \\ 3\cdot 2 + 0\cdot 0 + 1\cdot 1 & 3\cdot 1 + 0\cdot 3 + 1\cdot 0 \\ 1\cdot 2 + 1\cdot 0 + 2\cdot 1 & 1\cdot 1 + 1\cdot 3 + 2\cdot 0 \end{pmatrix} = \begin{pmatrix} 2 & 7 \\ 7 & 3 \\ 4 & 4 \end{pmatrix}$$

Properties of Matrix Operations

Addition:

- $A + B = B + A$ (commutative)
- $(A + B) + C = A + (B + C)$
- $A + O = A$ (zero matrix)
- $A + (-A) = O$

Multiplication:

- $(AB)C = A(BC)$ (associative)
- $A(B + C) = AB + AC$
- $AI = IA = A$
- $(AB)^T = B^T A^T$

2. Determinants

Definition — Determinant

The **determinant** is a scalar value computed from a square matrix. For a 3×3 matrix A:

$$\det(A) = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

Sarrus' Rule (3×3)

Write the first two columns to the right of A, then sum the three downward diagonals and subtract the three upward diagonals.

Example. $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$

$$\begin{aligned} \det(A) &= (1 \cdot 5 \cdot 9 + 2 \cdot 6 \cdot 7 + 3 \cdot 4 \cdot 8) - (3 \cdot 5 \cdot 7 + 1 \cdot 6 \cdot 8 + 2 \cdot 4 \cdot 9) \\ &= (45 + 84 + 96) - (105 + 48 + 72) = 225 - 225 = 0 \end{aligned}$$

Key Formula

Properties of Determinants:

- $\det(AB) = \det(A) \cdot \det(B)$
- $\det(A^T) = \det(A)$
- $\det(kA) = k^n \det(A)$ for $n \times n$ matrix
- Swapping two rows/columns changes the sign of the determinant
- If two rows are identical, $\det(A) = 0$
- $\det(A^{-1}) = 1/\det(A)$ (if A is invertible)

3. Singular and Non-Singular Matrices

Definition — Singular Matrix

A square matrix A is **singular** if $\det(A) = 0$. A singular matrix has no inverse.

Definition — Non-Singular Matrix

A square matrix A is **non-singular** (invertible) if $\det(A) \neq 0$. Its inverse is

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

where $\operatorname{adj}(A)$ is the adjugate (transpose of the cofactor matrix).

Comparison Table:

Property	**Singular**	**Non-Singular**
$\det(A)$	$= 0$	$\neq 0$
Inverse	Does not exist	Exists
Rank	$< n$	$= n$ (full rank)
System $Ax = b$	May have no/infinite solutions	Unique solution

Example. $A = \begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$: $\det(A) = 4 - 4 = 0 \to$ **Singular**.

$B = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$: $\det(B) = 12 - 2 = 10 \neq 0 \to$ **Non-Singular**.

4. Minor and Cofactor

Definition — Minor

The **minor** M_{ij} of element a_{ij} is the determinant of the submatrix obtained by deleting row i and column j.

Definition — Cofactor

The **cofactor** C_{ij} is the signed minor:

$$C_{ij} = (-1)^{i+j} M_{ij}$$

The sign pattern for a 3×3 matrix is:

$$\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

Example. For $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{pmatrix}$:

Minor M_{11}: delete row 1, col 1 $\Rightarrow$ det $\begin{pmatrix} 4 & 5 \\ 0 & 6 \end{pmatrix} = 24$

Minor M_{12}: delete row 1, col 2 $\Rightarrow$ det $\begin{pmatrix} 0 & 5 \\ 1 & 6 \end{pmatrix} = -5$

Minor M_{13}: delete row 1, col 3 $\Rightarrow$ det $\begin{pmatrix} 0 & 4 \\ 1 & 0 \end{pmatrix} = -4$

$$\det(A) = 1(24) - 2(-5) + 3(-4) = 24 + 10 - 12 = 22$$

Cofactor Expansion: $\det(A) = \sum_{j=1}^{n} a_{ij} C_{ij}$ along any row i, or $\sum_{i=1}^{n} a_{ij} C_{ij}$ along any column j.

5. Gauss Elimination (3×3)

Definition — Gauss Elimination

Gaussian Elimination reduces the augmented matrix $[A|b]$ to **upper triangular** (Row Echelon Form), then uses **back substitution** to solve.

Method — Steps

1. Write the augmented matrix $[A|b]$.
2. Use row operations to create zeros below each pivot (leading entry).
3. Achieve Row Echelon Form (upper triangular).
4. Back-substitute from the last equation upward.

Example. Solve:

$$x + 2y + z = 5, \quad 3x + 5y + z = 10, \quad 2x + 4y + z = 8$$

Augmented matrix and row operations:

$$\begin{pmatrix} 1 & 2 & 1 & 5 \\ 3 & 5 & 1 & 10 \\ 2 & 4 & 1 & 8 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - 3R_1} \begin{pmatrix} 1 & 2 & 1 & 5 \\ 0 & -1 & -2 & -5 \\ 2 & 4 & 1 & 8 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - 2R_1} \begin{pmatrix} 1 & 2 & 1 & 5 \\ 0 & -1 & -2 & -5 \\ 0 & 0 & -1 & -2 \end{pmatrix}$$

Back substitution: $z = 2$, $-y - 4 = -5 \Rightarrow y = 1$, $x + 2 + 2 = 5 \Rightarrow x = 1$.

$$\boxed{x = 1, \quad y = 1, \quad z = 2}$$

6. Gauss–Jordan Elimination (3×3)

Definition — Gauss–Jordan Elimination

Gauss–Jordan extends Gaussian elimination to produce **Reduced Row Echelon Form (RREF)**: each pivot is 1 and all other entries in that column are 0. No back substitution needed.

Example. Same system as above:

$$\begin{pmatrix} 1 & 2 & 1 & 5 \\ 0 & -1 & -2 & -5 \\ 0 & 0 & -1 & -2 \end{pmatrix} \xrightarrow{R_3 \leftarrow -R_3} \begin{pmatrix} 1 & 2 & 1 & 5 \\ 0 & -1 & -2 & -5 \\ 0 & 0 & 1 & 2 \end{pmatrix}$$

$$\xrightarrow{R_2 \leftarrow R_2 + 2R_3} \begin{pmatrix} 1 & 2 & 1 & 5 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{pmatrix} \xrightarrow{R_2 \leftarrow -R_2} \begin{pmatrix} 1 & 2 & 1 & 5 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{pmatrix}$$

$$\xrightarrow{R_1 \leftarrow R_1 - 2R_2 - R_3} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{pmatrix}$$

Result directly: $x = 1$, $y = 1$, $z = 2$.

Gauss vs Gauss–Jordan: Gaussian gives upper triangular + back substitution. Gauss–Jordan gives full RREF — solution read directly. Gauss–Jordan is used to find matrix inverses by augmenting with the identity $[A|I] \to [I|A^{-1}]$.

7. Cramer's Rule

Definition — Cramer's Rule

For a system $Ax = b$ with $\det(A) \neq 0$, each variable x_i is given by:

$$x_i = \frac{\det(A_i)}{\det(A)}$$

where A_i is the matrix A with its i-th column replaced by b.

Example. Solve:

$$2x + y - z = 3, \quad x - y + 2z = 1, \quad 3x + 2y + z = 4$$

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 1 & -1 & 2 \\ 3 & 2 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix}$$

$\det(A) = 2(-1-4) - 1(1-6) + (-1)(2+3) = -10 + 5 - 5 = -10$

$$A_1 = \begin{pmatrix} 3 & 1 & -1 \\ 1 & -1 & 2 \\ 4 & 2 & 1 \end{pmatrix} \Rightarrow \det(A_1) = 3(-1-4) - 1(1-8) + (-1)(2+4) = -15 + 7 - 6 = -14$$

$$A_2 = \begin{pmatrix} 2 & 3 & -1 \\ 1 & 1 & 2 \\ 3 & 4 & 1 \end{pmatrix} \Rightarrow \det(A_2) = 2(1-8) - 3(1-6) + (-1)(4-3) = -14 + 15 - 1 = 0$$

$$A_3 = \begin{pmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 3 & 2 & 4 \end{pmatrix} \Rightarrow \det(A_3) = 2(-4-2) - 1(4-3) + 3(2+3) = -12 - 1 + 15 = 2$$

$$\boxed{x = \frac{-14}{-10} = \frac{7}{5}, \quad y = \frac{0}{-10} = 0, \quad z = \frac{2}{-10} = -\frac{1}{5}}$$

Cramer's Rule is elegant for small systems but computationally expensive for large n. Use Gaussian elimination for $n \geq 4$.

8. Orthogonal and Orthonormal Sets

Definition — Orthogonal Set

A set of vectors $\{v_1, v_2, \ldots, v_k\}$ in $\mathbb{R}^n$ is **orthogonal** if every pair of distinct vectors is perpendicular:

$$v_i \cdot v_j = 0 \quad \text{for all } i \neq j$$

Definition — Orthonormal Set

An orthogonal set where each vector has unit length ($\|v_i\| = 1$) is called **orthonormal**. To convert an orthogonal set to orthonormal, normalize each vector:

$$\hat{v}_i = \frac{v_i}{\|v_i\|}$$

Example. Check if $\{v_1, v_2, v_3\}$ is orthogonal:

$$v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$v_1 \cdot v_2 = 0$, $v_1 \cdot v_3 = 0$, $v_2 \cdot v_3 = 0 \to$ **Orthogonal**.
$\|v_1\| = \|v_2\| = \|v_3\| = 1 \to$ also **Orthonormal** (standard basis).

Non-trivial Example.

$$u_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad u_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \quad u_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$u_1 \cdot u_2 = -1 + 1 + 0 = 0\checkmark$, $u_1 \cdot u_3 = 0\checkmark$, $u_2 \cdot u_3 = 0\checkmark \to$ Orthogonal.

Normalize: $\hat{u}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, $\hat{u}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$, $\hat{u}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \to$ Orthonormal.

Theorem — Gram–Schmidt Process

Given a linearly independent set $\{x_1, \ldots, x_k\}$, produce an orthogonal set $\{v_1, \ldots, v_k\}$:

$$v_1 = x_1, \qquad v_k = x_k - \sum_{j=1}^{k-1} \frac{x_k \cdot v_j}{\|v_j\|^2} v_j$$

Then normalize to get an orthonormal set.

9. Projection of a Vector

Definition — Projection

The **orthogonal projection** of vector u onto vector v is:

$$\text{proj}_v\, u = \frac{u \cdot v}{\|v\|^2}\, v = \frac{u \cdot v}{v \cdot v}\, v$$

The **scalar projection** (component) of u onto v is:

$$\text{comp}_v\, u = \frac{u \cdot v}{\|v\|}$$

Example 1. Project $u = (3, 4)$ onto $v = (1, 0)$:

$$\text{proj}_v\, u = \frac{(3)(1) + (4)(0)}{1^2 + 0^2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$$

Example 2. Project $u = (2, 1, 3)$ onto $v = (1, 2, 2)$:

$$u \cdot v = 2 + 2 + 6 = 10, \quad \|v\|^2 = 1 + 4 + 4 = 9$$

$$\text{proj}_v\, u = \frac{10}{9} \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 10/9 \\ 20/9 \\ 20/9 \end{pmatrix}$$

Projection onto a subspace spanned by orthonormal basis $\{\hat{u}_1, \ldots, \hat{u}_k\}$:

$$\text{proj}_W\, x = (x \cdot \hat{u}_1)\hat{u}_1 + (x \cdot \hat{u}_2)\hat{u}_2 + \cdots + (x \cdot \hat{u}_k)\hat{u}_k$$

Key Formula

Projection Matrix. The projection matrix onto the column space of A (with linearly independent columns) is:

$$P = A(A^T A)^{-1} A^T$$

Then $\text{proj}_{\text{col}(A)}\, b = Pb$.

10. Group Theory

Definition — Group

A **group** is a set G together with a binary operation $*$ satisfying four axioms:

1. **Closure:** $\forall\, a, b \in G,\ a * b \in G$
2. **Associativity:** $\forall\, a, b, c \in G,\ (a * b) * c = a * (b * c)$
3. **Identity:** $\exists\, e \in G$ such that $a * e = e * a = a$
4. **Inverse:** $\forall\, a \in G,\ \exists\, a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$

If also $a * b = b * a$ for all $a, b \in G$, the group is **Abelian (commutative)**.

Common Examples of Groups

Group	Set	Operation	Abelian?
$(\mathbb{Z}, +)$	Integers	Addition	Yes
$(\mathbb{Q} \setminus \{0\}, \times)$	Nonzero rationals	Multiplication	Yes
$(GL_n(\mathbb{R}), \cdot)$	Invertible $n \times n$ matrices	Matrix mult.	No ($n \geq 2$)
$(\mathbb{Z}_n, +)$	$\{0, 1, \ldots, n-1\}$	Addition mod n	Yes
S_n	Permutations of n elements	Composition	No ($n \geq 3$)

Subgroup

Definition — Subgroup

A subset $H \subseteq G$ is a **subgroup** of G if H itself forms a group under the same operation. Equivalently, H is a subgroup iff:

- $e \in H$
- $a, b \in H \Rightarrow a * b \in H$ (closure)
- $a \in H \Rightarrow a^{-1} \in H$

Example. $(\mathbb{Z}, +)$ is a group. $H = 2\mathbb{Z} = \{\ldots, -4, -2, 0, 2, 4, \ldots\}$ (even integers) is a subgroup: $0 \in H$, sum of evens is even, negative of even is even.

Order of a Group and Element

Order of group $|G|$: number of elements.
Order of element a: smallest positive integer n with $a^n = e$. If no such n, order is infinite.

Example. In $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$ under addition mod 6:

$\operatorname{ord}(1) = 6$ (since $6 \cdot 1 = 6 \equiv 0$), $\operatorname{ord}(2) = 3$, $\operatorname{ord}(3) = 2$, $\operatorname{ord}(0) = 1$.

Theorem — Lagrange's Theorem

If H is a subgroup of a finite group G, then $|H|$ divides $|G|$.

Quick Reference

Linear Algebra — Cheat Sheet

DETERMINANTS & INVERSES

$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$ $A^{-1} = \frac{1}{\det A}\text{adj}(A)$

$\det(AB) = \det A \cdot \det B$ $\det(A^T) = \det A$

Singular: $\det A = 0$ Non-singular: $\det A \neq 0$

SOLVING LINEAR SYSTEMS

Gauss Elimination: $[A|b] \to$ upper triangular $\to$ back-sub

Gauss–Jordan: $[A|b] \to$ RREF $\to$ read solution

Cramer's Rule: $x_i = \frac{\det(A_i)}{\det(A)}$ (replace col i with b)

Matrix Inverse: $[A|I] \xrightarrow{\text{RREF}} [I|A^{-1}]$

VECTORS & PROJECTIONS

$\text{proj}_v u = \frac{u \cdot v}{\|v\|^2} v$ $\text{comp}_v u = \frac{u \cdot v}{\|v\|}$

Orthogonal: $u \cdot v = 0$ Normalize: $\hat{v} = \frac{v}{\|v\|}$

Projection matrix: $P = A(A^T A)^{-1} A^T$

GROUP THEORY

Group $(G, *)$: Closure, Associativity, Identity e, Inverse a^{-1}

Abelian: $a * b = b * a$ Subgroup: subset closed under $*$ and inverses

Lagrange: $|H|$ divides $|G|$ $\text{ord}(a)$: smallest n with $a^n = e$

For feedback or to request additional topics, contact:

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